

Three ways to count a 6D gauge field. Two of them agree.

KM25 eq. (62), as printed

4

A_μ

+

1

surviving scalar

=

5

$\times$

covariant 6D (HNT)

4

A_μ

-

2

ghosts

+

1

A_5

+

1

A_6

=

4

$\checkmark$

physical degrees

3

massive vector

+

1

orthogonal scalar

=

4

$\checkmark$

a 6D gauge field has $D - 2 = 4$

What that does to the parity of 8D

the row that mixes both sectors is $2a + 6b$, over $n_+ = 2$ and $n_- = 6$ pairs

KM25 as printed, $(a, b) = (2, \frac{1}{2})$

4

periodic $2a$

+

3

antiperiodic $6b$

=

7

$\longrightarrow$

$8D = -27$

ODD

Theorem 1 applies

candidate count, $(a, b) = (\frac{3}{2}, \frac{1}{2})$

3

periodic $2a$

+

3

antiperiodic $6b$

=

6

$\longrightarrow$

$8D = -18$

EVEN

Theorem 1 says nothing here

The antiperiodic part is 3 either way. What moves is the periodic one.